

SHILPA REDDY

Data Engineer

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PROFILE SUMMARY

- Data Engineer with 5+ years of experience building and operating scalable data platforms across Azure and AWS, supporting real-time analytics, reporting, and AI/ML use cases.
- Experience designing end-to-end batch and streaming data pipelines for high-volume transactional, event-driven, and time-series data in production environments.
- Strong programming background in Python, Scala, Java, and SQL, developing reliable ETL/ELT pipelines, ingestion services, and distributed data processing jobs.
- Hands-on cloud data engineering experience on Azure (Azure Data Factory, Databricks, ADLS, Synapse) and AWS (EMR, Glue, Lambda, S3, EC2).
- Experience with big data and streaming frameworks including Spark, PySpark, Kafka, Hive, and Spark Structured Streaming for large-scale data processing.
- Implemented Lakehouse architectures (Bronze–Silver–Gold) using Databricks and Delta Lake to enable consistent data modeling for BI, analytics, and ML workloads.
- Strong production ownership with on-call support experience, debugging pipeline failures, streaming lag, and data quality issues, and delivering permanent fixes under strict SLAs.
- Enabled AI and ML workflows by building feature-ready datasets and integrating Azure ML and MLflow for model training, versioning, automated batch inference, and continuous retraining.
- Experience in workflow orchestration and automation using Azure Data Factory, AWS Glue, and Apache Airflow to manage complex data pipelines.
- Collaborative engineer with experience working in Agile/Scrum environments, partnering with ML engineers, analysts, and business teams to deliver business-impacting data solutions.

TECHNICAL SKILLS

Category	Technologies
Cloud Platforms	Microsoft Azure: Azure Data Factory, Azure Databricks, Azure Data Lake Storage (ADLS), Azure Synapse Analytics (SQL), Azure Functions, Azure Logic Apps, Azure Key Vault, Azure ML Studio AWS: Amazon EMR, Amazon S3, AWS Glue, AWS Lambda, AWS Step Functions, Amazon EC2, Amazon EKS, CloudWatch, IAM
Databases & Storage	Snowflake, MySQL, Oracle, MongoDB, HBase, Delta Lake, Hive
Programming Languages	Python, SQL, Scala, Java
Big Data & Streaming	Apache Spark, PySpark, Kafka, Spark Structured Streaming, Hadoop
Orchestration & Automation	Apache Airflow, Azure Data Factory, AWS Glue
MLOps & ML Enablement	MLflow, Azure ML
CI/CD & Containers	Jenkins, Git, Docker, Kubernetes
BI & Analytics	Power BI, Pandas, NumPy

EDUCATION

Indiana Wesleyan University, Indianapolis, USA

- Masters / Computer Science - Aug 2022 to May 2024

RELEVANT EXPERIENCE

Allstate Insurance, Northbrook, IL

Data Engineer

Oct 2024 – Present

- Worked on a real-time risk and claims analytics platform supporting underwriting and analytics teams with large-scale policy, claims, and telematics data.
- Designed and maintained end-to-end ETL pipelines using Azure Data Factory and Databricks, ingesting batch and streaming data into the enterprise data lake.
- Built and optimized PySpark pipelines processing millions of records daily, handling schema evolution, late-arriving data, and data quality edge cases.
- Implemented a Lakehouse architecture (Bronze–Silver–Gold) on Azure Blob Storage and Delta Lake to enable reliable BI, analytics, and ML consumption.
- Developed real-time streaming pipelines using Kafka and Spark Structured Streaming to process driver behavior and claims events with low latency.
- Owned production support for data pipelines, resolving ADF failures, Databricks job errors, Kafka lag, and data inconsistencies to meet strict SLAs.
- Automated ingestion from on-prem MySQL to Azure Synapse and Snowflake, implementing reconciliation checks, validations, and post-load transformations.
- Used Pandas, NumPy, and Matplotlib for data validation, anomaly detection, and exploratory analysis, reducing downstream data issues by ~25%.
- Enabled ML and AI workflows by preparing feature datasets and integrating Azure ML and MLflow for model training, versioning, and batch inference.
- Embedded model execution into ADF pipelines to support automated scoring and continuous retraining of customer risk models.
- Built CI/CD pipelines using Jenkins and Git and implemented infrastructure automation with Terraform to support reliable, repeatable deployments.
- Collaborated in an Agile environment with data architects, ML engineers, analysts, and QA teams to deliver and improve production data platforms.

Environment: Azure Data Factory, Azure Databricks, Azure Data Lake Storage (ADLS), Delta Lake, Azure Synapse Analytics, Snowflake, Kafka, Spark Structured Streaming, PySpark, Pandas, NumPy, Matplotlib, Azure ML Studio, MLflow, Azure Functions, Azure Logic Apps, Jenkins, Git, Terraform, Docker, Kubernetes, MySQL, Power BI, Azure AD, Azure Key Vault, Agile/Scrum.

ACI Worldwide, Naples, Florida, USA

Jul 2023 - Sept 2024

Data Engineer

- Owned data engineering work for high-volume payment processing and settlement platforms, enabling near real-time fraud detection, reconciliation, and compliance analytics.
- Designed and implemented low-level Java APIs to ingest large volumes of structured and semi-structured transaction data into MongoDB and HBase, supporting high-throughput writes and low-latency downstream access.
- Built and maintained Python-based ETL pipelines to ingest, cleanse, and transform transactional and settlement data from multiple payment gateways for analytics and operational reporting.
- Orchestrated batch and streaming ingestion workflows using AWS Glue, Lambda, and Step Functions, improving pipeline reliability and reducing manual intervention in production.
- Developed distributed Spark applications to aggregate and reconcile authorization, clearing, and settlement records, directly supporting fraud monitoring and financial reporting workflows.
- Designed and optimized MongoDB data models, managing incremental ingestion, schema evolution, and consistency checks across large merchant and transaction datasets.
- Built Kafka streaming pipelines to capture near real-time transaction events and persist them in MongoDB, enabling timely fraud signal availability for downstream analytics.
- Took ownership of production support and on-call issues, analyzing CloudWatch and ELK logs to resolve Spark failures, Kafka lag, and data inconsistencies during peak transaction loads.
- Containerized and deployed Spark workloads using Docker and Kubernetes, improving fault tolerance and reducing recovery time during high-volume processing windows.
- Provisioned and managed AWS EC2 infrastructure and implemented Terraform-based automation to ensure scalable, consistent, and repeatable environment setups.
- Built and maintained CI/CD pipelines using Jenkins and Git, integrating automated testing and static analysis to improve deployment reliability and code quality.
- Collaborated in an Agile environment with finance, compliance, and business teams to deliver analytics-ready datasets while meeting strict SLA and regulatory requirements.

Environment: AWS EMR, Amazon S3, AWS Glue, AWS Lambda, AWS Step Functions, Amazon EC2, Amazon EKS, Kafka, Apache Spark, Scala, Python, MongoDB, HBase, Jenkins, Git, Terraform, Docker, Kubernetes, CloudWatch, ELK Stack (Elasticsearch, Logstash, Kibana), Power BI, Agile/Scrum.

Dupont & IFF Merger, Hyderabad, India
Data Engineer

Nov 2019 - Jul 2022

- Contributed to large-scale batch and streaming data platforms supporting enterprise analytics and reporting during the DuPont–IFF merger integration.
- Built and maintained batch and real-time data processing pipelines using Spark and Spark Streaming on Amazon EMR to handle high-volume datasets.
- Developed PySpark and Spark SQL pipelines to transform multi-format data (CSV, JSON, Parquet) stored in Amazon S3 into analytics-ready datasets.
- Implemented large-scale data cleansing and preprocessing workflows using Hadoop and Hive on EMR clusters to improve data quality and consistency.
- Managed Spark workloads on Amazon EMR and deployed containerized data processing jobs using Amazon EKS to improve scalability, isolation, and resource utilization.
- Built reusable data ingestion frameworks using AWS Glue, including secure batch file ingestion from external vendors via SFTP.
- Orchestrated end-to-end workflows using Apache Airflow to schedule, monitor, and manage Spark and Glue jobs with dependency handling and failure recovery.
- Used Sqoop to migrate structured data from on-prem Oracle systems into Amazon S3 for downstream analytics and reporting.
- Monitored and supported production pipelines using Amazon CloudWatch, identifying failures and performance bottlenecks in EMR, Glue, and EKS workloads.
- Created business and cost analysis dashboards using Amazon QuickSight and Tableau based on curated analytics datasets.

Environment: AWS EMR, Amazon S3, AWS Glue, Apache Airflow, Apache Spark, Spark Streaming, PySpark, Spark SQL, Hadoop, Hive, Sqoop, Amazon EKS, CloudWatch, Oracle, QuickSight, Tableau, Python, Scala, Git, Agile/Scrum.